

It's easy to see that $(2K+1)K_2(2K+2)+2$ is an integer.

By defn. of an odd integer, in defn 1, $(n+5)-n$ is odd.

By defn 1, an integer can't be both even and odd at the same time.

$\therefore$ This is a contradiction $\square$

$\therefore \neg q$ is False. By defn. of AND, $(p \wedge \neg q)$ is False. By defn. of conditional,

$(p \wedge \neg q) \rightarrow F$ is True. $\therefore p \rightarrow \neg q$ is True

$\therefore$ If $n+5$ is an odd integer, then n is an even integer. $\square$

2b) Prove that if n is an integer and $3n+2$ is even, then n is even using

a) a proof by contraposition b) a proof by contradiction.

Solution: p : $3n+2$ is an even integer. q : n is an even integer.

a) We shall prove $(p \rightarrow q)$ is True by contraposition.

$\therefore$ We shall prove $(\neg q \rightarrow \neg p)$ is True.

$\neg p$: $(3n+2)$ is not even, i.e. $(3n+2)$ is an odd integer (By Defn. 1)

$\neg q$: n is not even, i.e. n is an odd integer. (By Defn 1)

Let, $\neg q$ be True. $\therefore n$ is an arbitrary odd integer. By Defn. 1, there exists an integer K such that $n = 2K+1$.

$\therefore 3n+2 = 3(2K+1)+2 = 6K+3+2 = (6K+4)+1 = 2(3K+2)+1$

$\therefore K$ is an integer, $(3K+2)$ is an integer. $\therefore$ By Defn 1, $(3n+2)$ is an odd integer. $\therefore \neg p$ is True. By defn. of conditional, $(\neg q \rightarrow \neg p)$ is True.

$\therefore p \rightarrow q$ is True $\square$